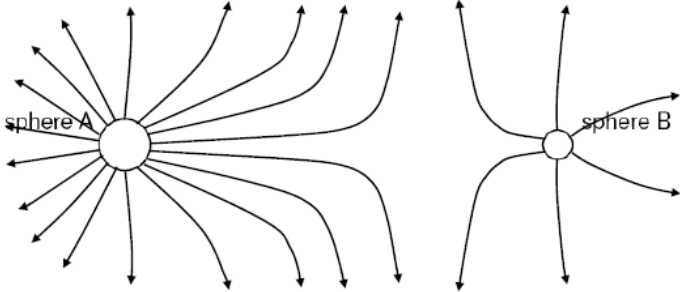


3	(a)	The <i>electric potential</i> at a point in an electric field is defined as the work done per unit positive charge to bring a test charge from infinity to the point by an external agent.		B1
	(b)	Diagram  Deduct 1 mark if any of the following is not shown. • Correct field line direction and shape. • At least more field lines radiating out of sphere A than sphere B. • Location of neutral point nearer to sphere B.		A1 A2 A3
	(c)	(i)	$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ $V_o = 2 \times \frac{1}{4\pi\epsilon_0} \frac{(-2.0 \times 10^{-9})}{(0.050)} + 2 \times \frac{1}{4\pi\epsilon_0} \frac{(-2.0 \times 10^{-9})}{\sqrt{0.050^2 + 0.100^2}}$ $= -1041V \approx -1000V \text{ (2s.f)}$	C1 A1
		(ii)	At infinity, Electric potential = 0 V By conservation of energy, moving from infinity to A, Gain of EPE = loss of KE $U_f - U_i = KE_i - KE_f$ $-1040(-1.6 \times 10^{-19}) - 0 = \frac{1}{2}(9.11 \times 10^{-31})v_{\min}^2 - 0$ $v_{\min} = 1.9 \times 10^7 \text{ m s}^{-1}$	B1 C1 A1